

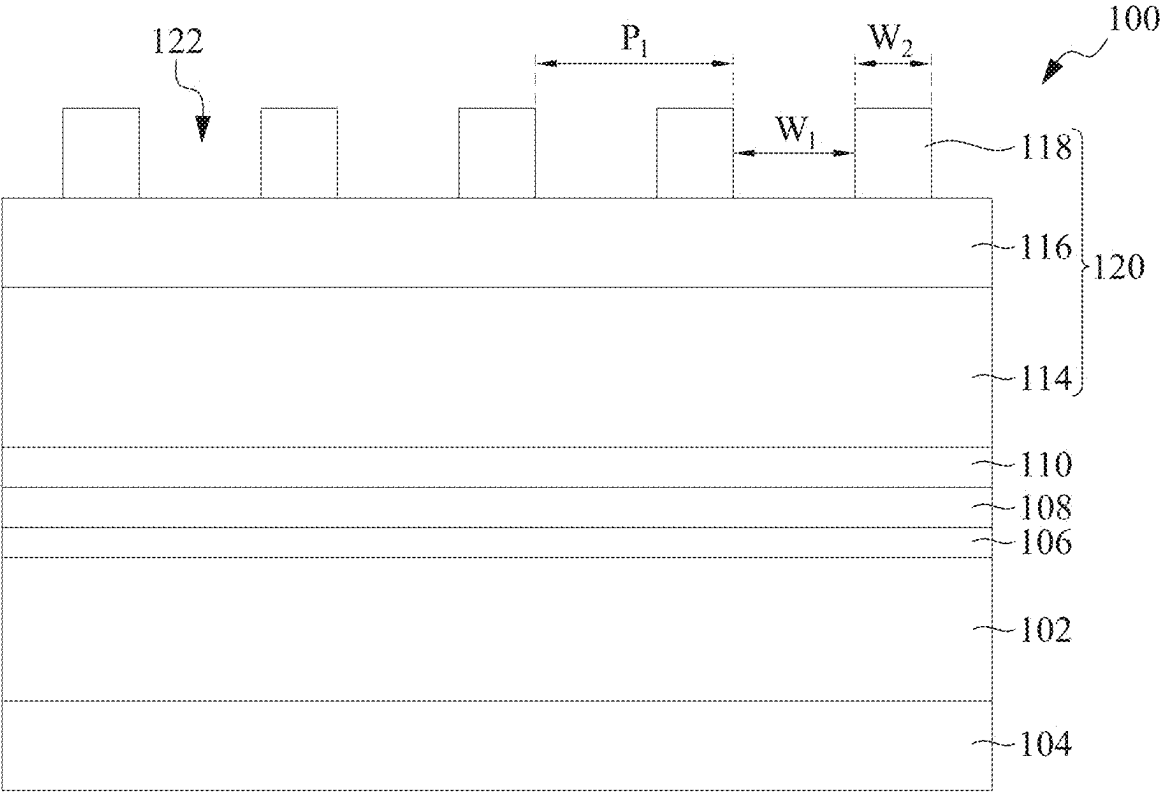


Fig. 1

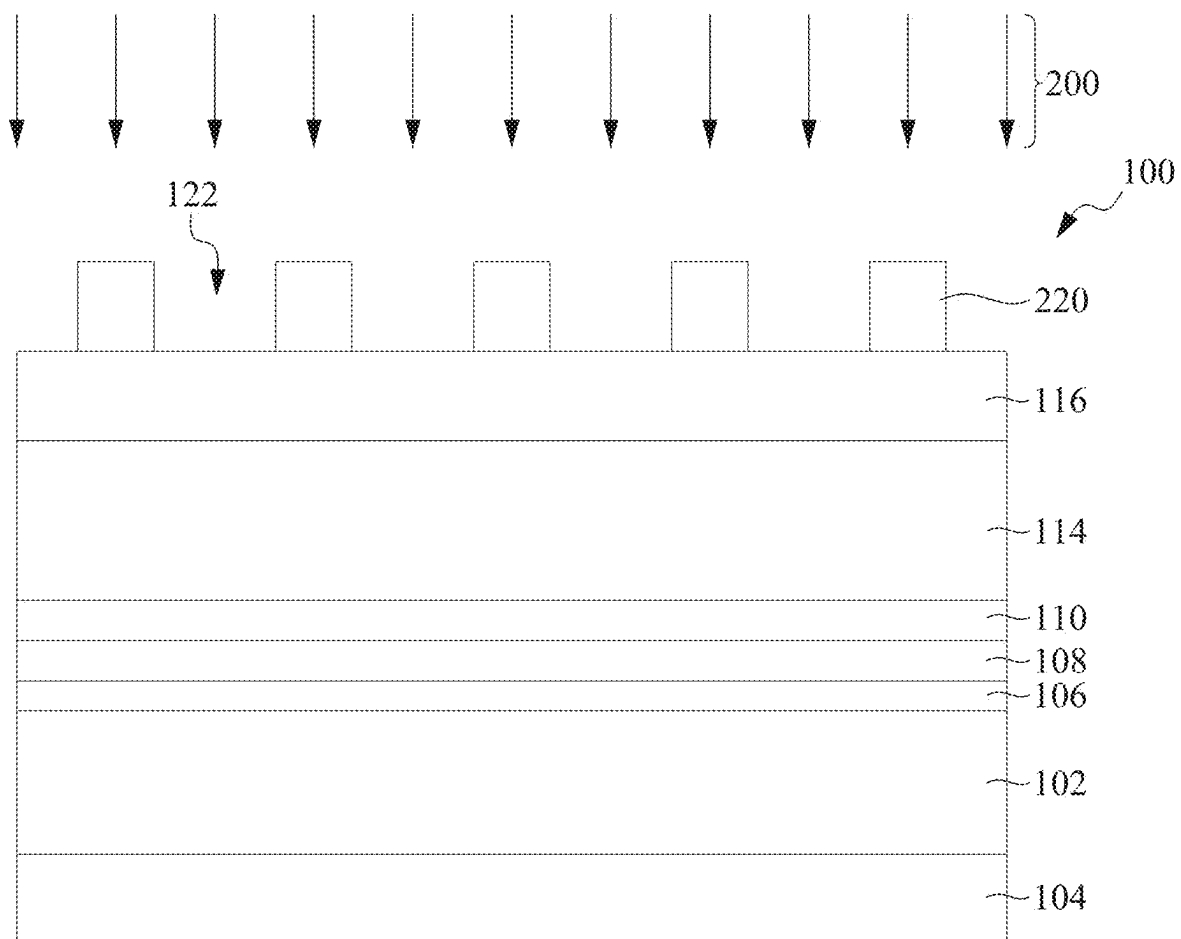


Fig. 2

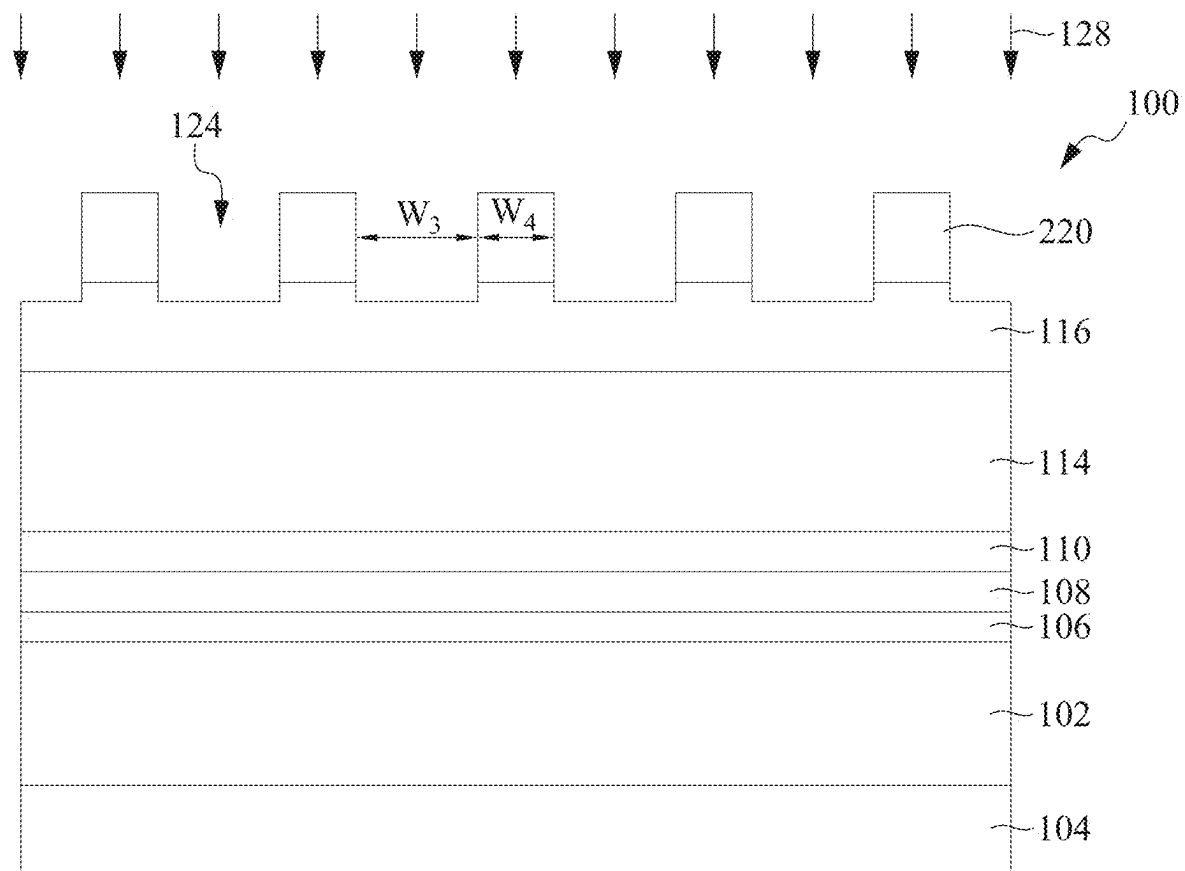


Fig. 3

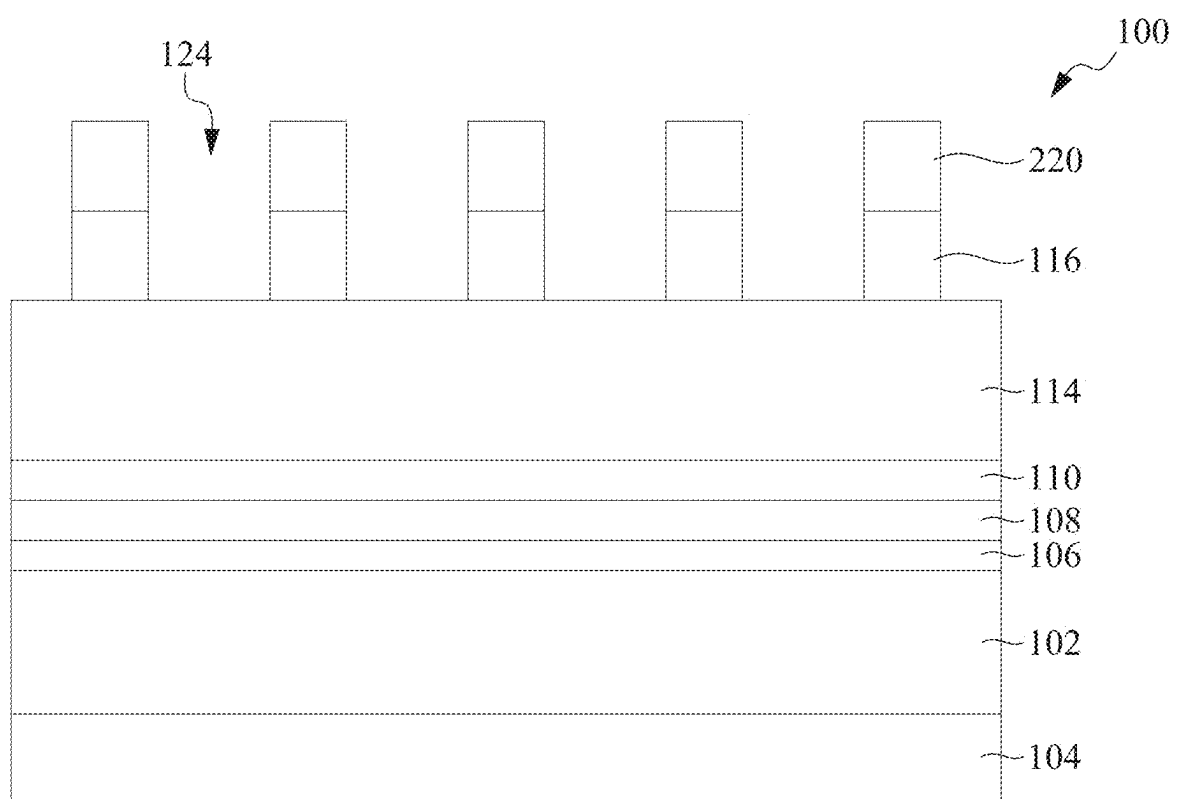


Fig. 4

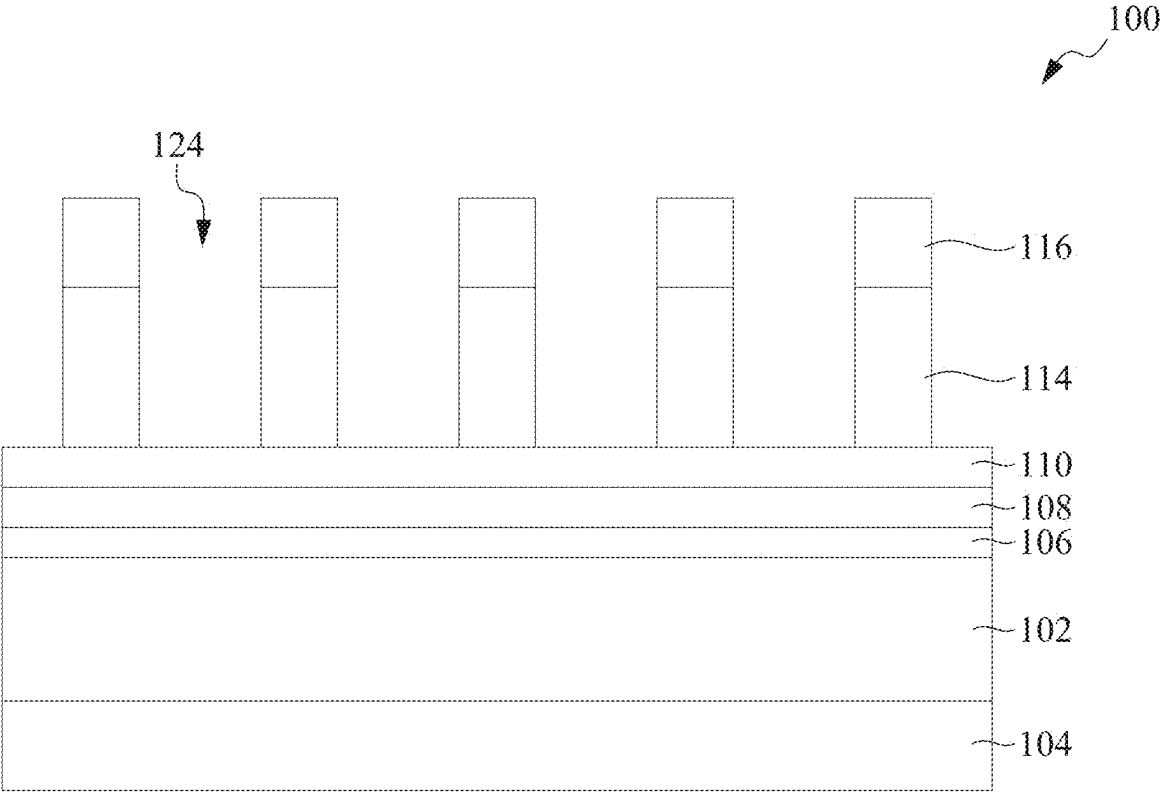


Fig. 5

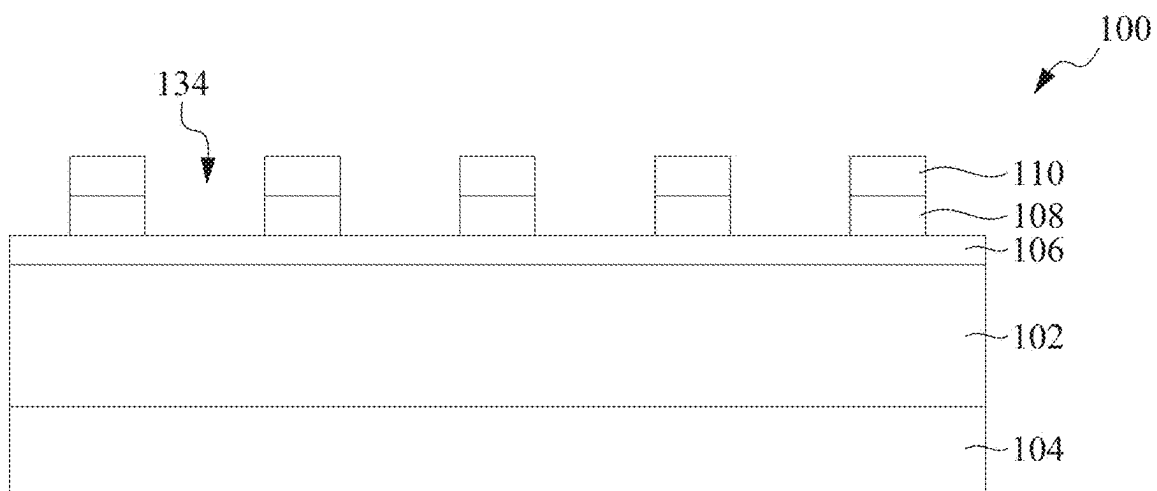


Fig. 6

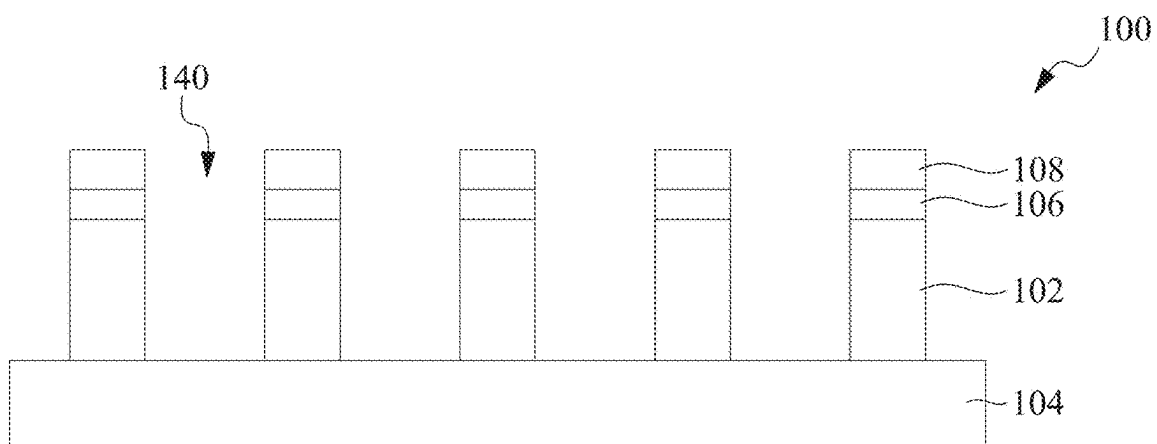


Fig. 7

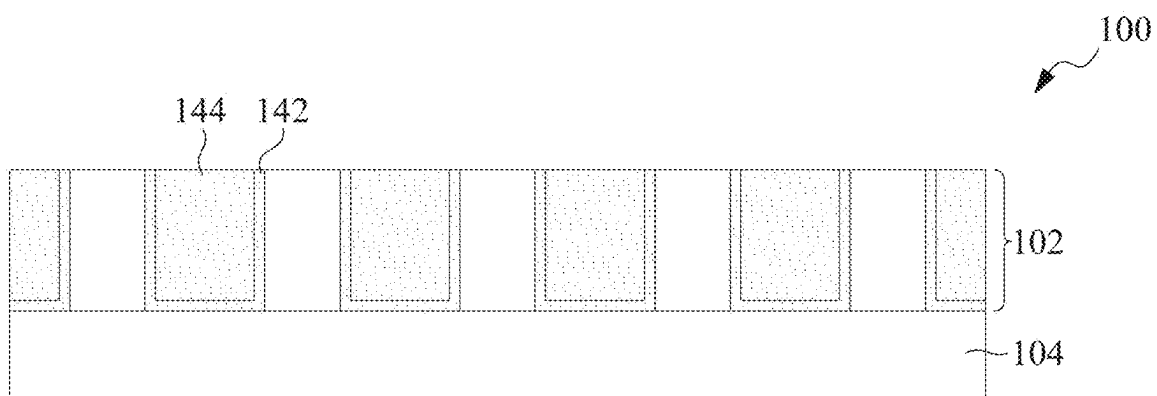


Fig. 8

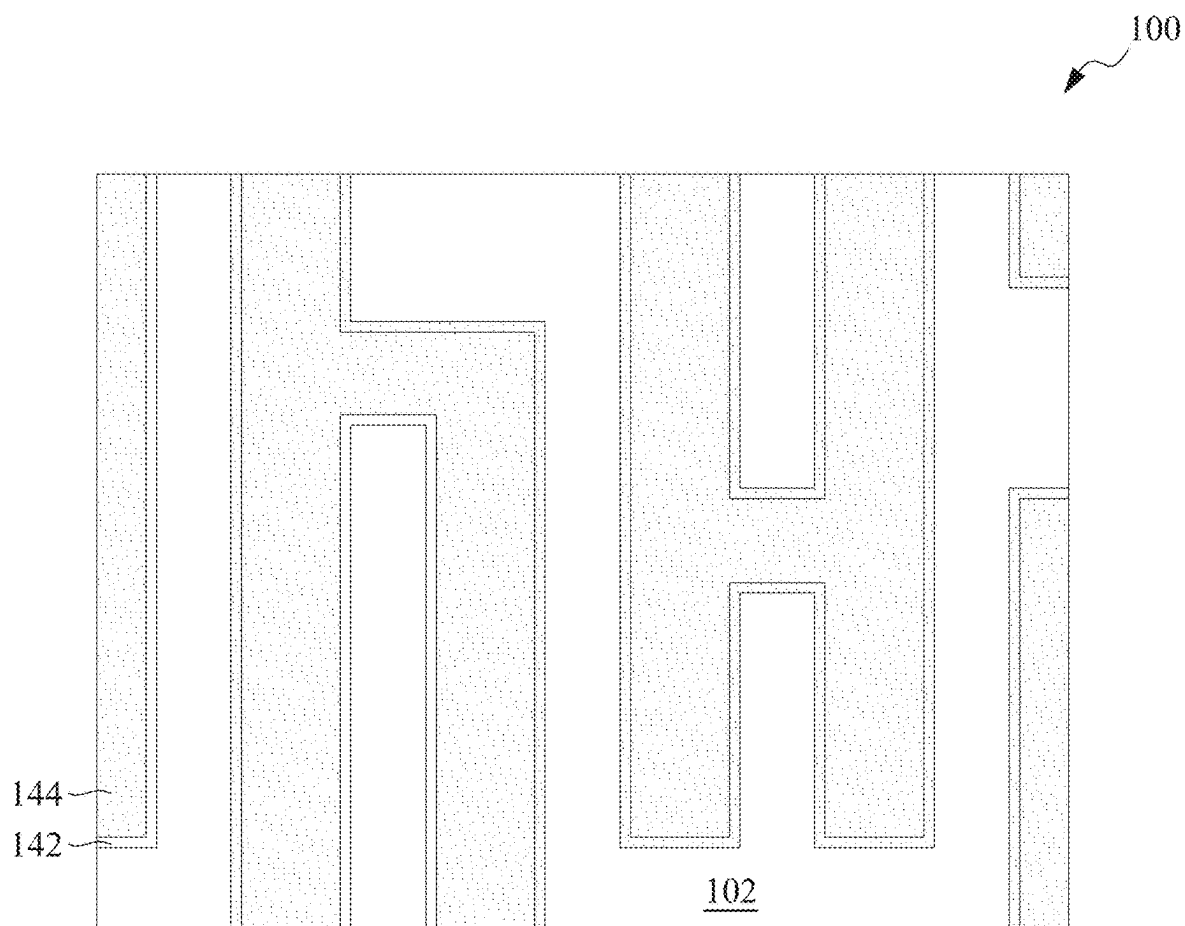


Fig. 9

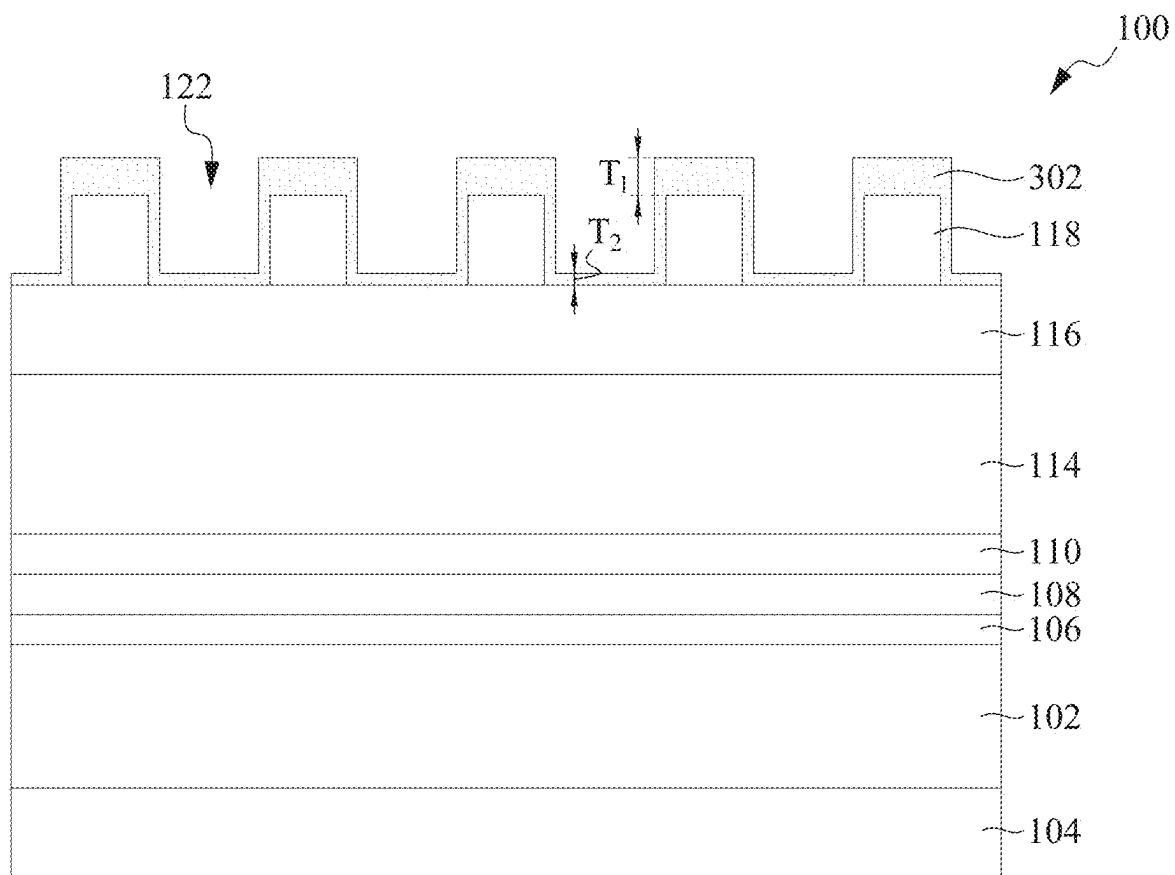


Fig. 10

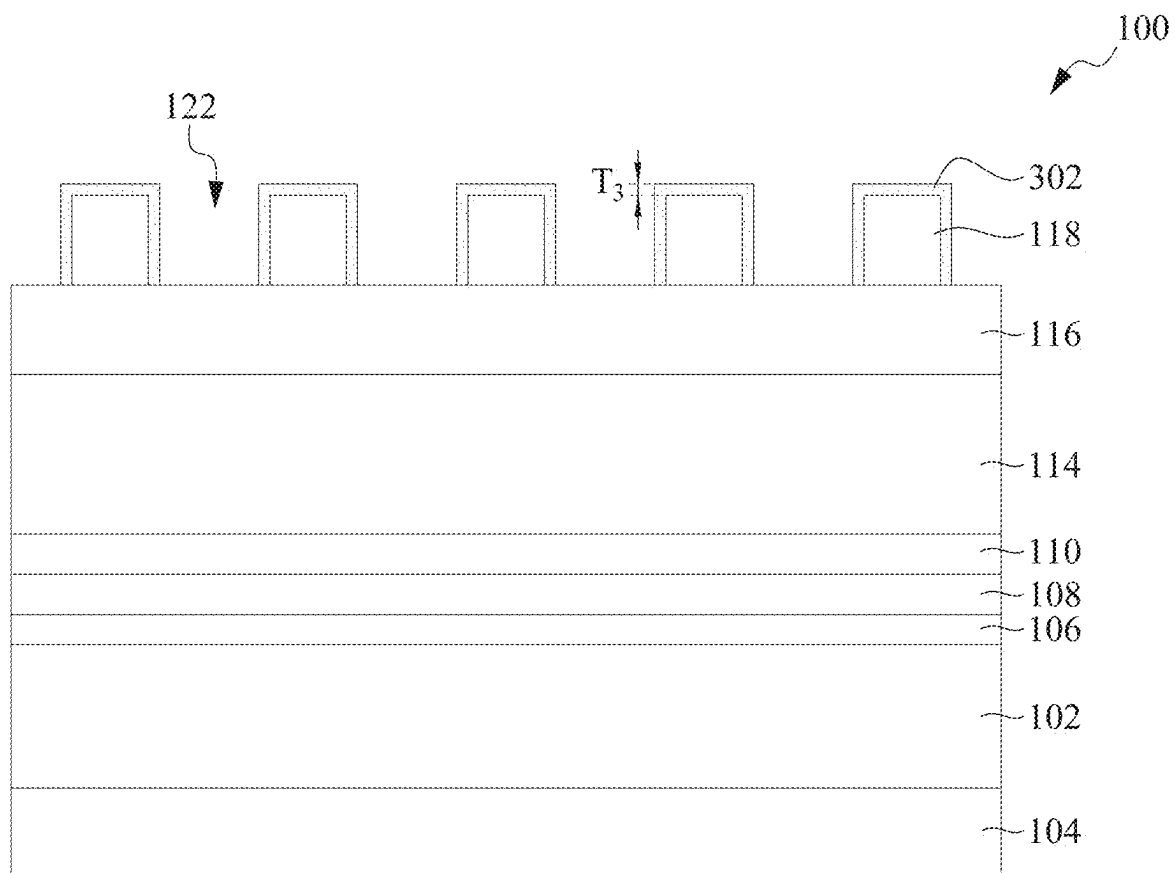


Fig. 11

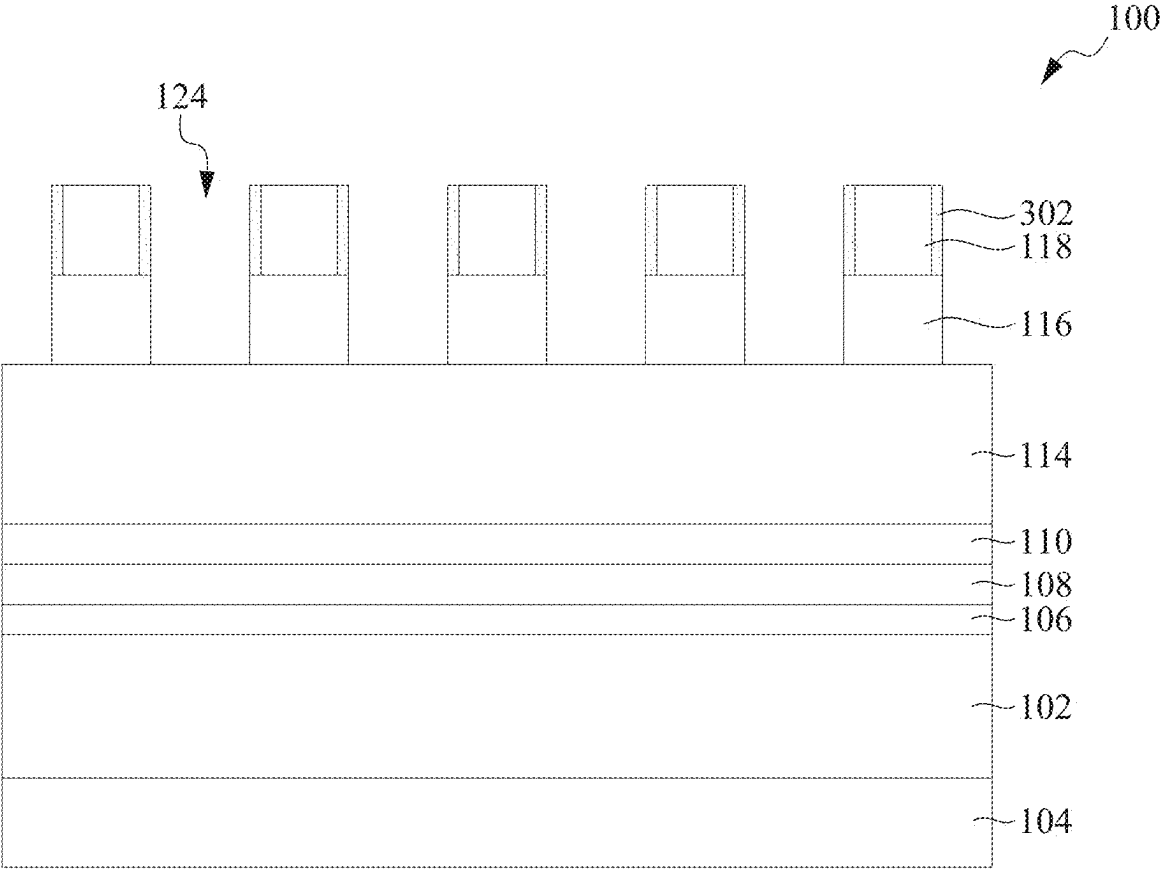


Fig. 12

METHOD FOR FORMING AND USING MASK

PRIORITY CLAIM AND CROSS-REFERENCE

[0001] This application is a continuation of U.S. patent application Ser. No. 18/401,800, filed on Jan. 2, 2024, which is a continuation of U.S. patent application Ser. No. 17/388,209, filed on Jul. 29, 2021, now U.S. Pat. No. 11,887,851 issued Jan. 30, 2024, each application is hereby incorporated herein by reference.

BACKGROUND

[0002] With the increasing down-scaling of semiconductor devices, various processing techniques (e.g., photolithography) are adapted to allow for the manufacture of devices with increasingly smaller dimensions. For example, as the density of gates increases, the manufacturing processes of various features in the device (e.g., overlying interconnect features) are adapted to be compatible with the down-scaling of device features as a whole. However, as semiconductor processes have increasingly smaller process windows, the manufacture of these devices have approached and even surpassed the theoretical limits of photolithography equipment. As semiconductor devices continue to shrink, the spacing desired between elements (i.e., the pitch) of a device is less than the pitch that can be manufactured using traditional optical masks and photolithography equipment.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 illustrates a cross-sectional view of an intermediary stage of manufacturing a semiconductor device in accordance with some embodiments.

[0005] FIGS. 2-8 illustrate cross-sectional views of various intermediary stages of manufacturing a semiconductor device in accordance with some embodiments.

[0006] FIG. 9 illustrates a top down view of an intermediary stage of manufacturing a semiconductor device in accordance with some embodiments.

[0007] FIGS. 10-12 illustrate cross-sectional views of various intermediary stages of manufacturing a semiconductor device in accordance with some embodiments.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In

addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] A semiconductor device and method are provided in accordance with some embodiments. In particular, methods of protecting a profile of a patterned photoresist layer are provided, including treating the photoresist with a treatment process (e.g., a plasma treatment) and forming a film over the photoresist. Protecting the profile of the patterned photoresist layer can lead to a transfer of the pattern of the patterned photoresist to lower layers with reduced defects, which may reduce line width roughness (LWR) of subsequently formed conductive features in a target layer, such as metal lines. Smoother profiles of the conductive features may decrease sheet resistance and thus improve device performance.

[0011] FIGS. 1 through 8 illustrate cross-sectional views of intermediate stages in the formation of features in a target layer 102 on a semiconductor device 100, in accordance with various embodiments. The target layer 102 is a layer in which a plurality of patterns is to be formed in accordance with embodiments of the present disclosure. In some embodiments, semiconductor device 100 is processed as part of a larger wafer. In such embodiments, after various features of the semiconductor device 100 is formed (e.g., active devices, interconnect structures, and the like), a singulation process may be applied to scribe line regions of the wafer in order to separate individual semiconductor dies from the wafer (also referred to as singulation).

[0012] In some embodiments, the target layer 102 is an inter-metal dielectric (IMD) layer. In such embodiments, the target layer 102 comprises a low-k dielectric material having a dielectric constant (k value) lower than 3.8, lower than 3.0, or lower than 2.5, for example. In some embodiments, target layer 102 is an IMD layer comprising high-k dielectric material having a k value higher than 3.8. Openings may be patterned in the target layer 102 with the embodiment processes, and conductive lines and/or vias may be formed in the openings as described below.

[0013] In some embodiments, the target layer 102 is a semiconductor substrate. The semiconductor substrate may be formed of a semiconductor material such as silicon, silicon germanium, or the like. In some embodiments, the semiconductor substrate is a crystalline semiconductor substrate such as a crystalline silicon substrate, a crystalline silicon carbon substrate, a crystalline silicon germanium substrate, a III-V compound semiconductor substrate, or the like. The semiconductor substrate may be patterned with an embodiment process to define semiconductor fins, and subsequent process steps may be used to form shallow trench isolation (STI) regions in the substrate. The semiconductor

finns may protrude from between the formed STI regions. Source/drain regions may be formed in the semiconductor fins, and gate dielectric and electrode layers may be formed over channels regions of the fins, thereby forming semiconductor devices such as fin field effect transistors (finFETs), nanostructure field effect transistors (nano-FETs), or the like.

[0014] In some embodiments, the target layer **102** is a conductive layer, such as a metal layer or a polysilicon layer, which is blanket deposited. Embodiment patterning processes may be applied to the target layer **102** in order to pattern semiconductor gates and/or dummy gates of transistors (e.g., finFETs, nano-FETs, or the like). By using embodiment processes to pattern a conductive target layer **102**, spacing between adjacent gates may be reduced and gate density may be increased.

[0015] In FIG. 1, a film stack including the target layer **102** is formed in semiconductor device **100**. In some embodiments, the target layer **102** may be formed over a semiconductor substrate **104**. The semiconductor substrate **104** may be formed of a semiconductor material such as silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate **104** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. Devices (not illustrated), such as transistors, diodes, capacitors, resistors, etc., may be formed in and/or on an active surface of semiconductor substrate **104**. In other embodiments where the target layer **102** is a semiconductor substrate used to form finFETs, the semiconductor substrate **104** may be omitted.

[0016] Although FIG. 1 illustrates target layer **102** being in physical contact with semiconductor substrate **104**, any number of intervening layers may be disposed between target layer **102** and semiconductor substrate **104**. Such intervening layers may include an inter-layer dielectric (ILD) layer comprising a low-k dielectric and having contact plugs formed therein, other IMD layers having conductive lines and/or vias formed therein, one or more intermediary layers (e.g., etch stop layers, adhesion layers, etc.), combinations thereof, and the like. For example, an optional etch stop layer (not illustrated) may be disposed directly under the target layer **102**. The etch stop layer may act as a stop for an etching process subsequently performed on the target layer **102**. The material and process used to form the etch stop layer may depend on the material of the target layer **102**. In some embodiments, the etch stop layer may be formed of silicon nitride, SiON, SiCON, SiC, SiOC, SiC_xN_y, SiO_x, other dielectrics, combinations thereof, or the like, and may be formed by plasma enhanced chemical vapor deposition (PECVD), low pressure CVD (LPCVD), plasma vapor deposition (PVD), or the like.

[0017] The film stack further includes an anti-reflective coating (ARC) **106** formed over the target layer **102**. The ARC **106** aids in the exposure and focus of overlying photoresist layers (discussed below) during patterning of the photoresist layers. In some embodiments, the ARC **106** may be formed from SiON, silicon carbide, materials doped with oxygen (O) and nitrogen (N), or the like. In some embodi-

ments, the ARC **106** is substantially free from nitrogen, and may be formed from an oxide. In such embodiments, the ARC **106** may be also referred to as a nitrogen-free ARC (NFARC). The ARC **106** may be formed by Plasma Enhance Chemical Vapor Deposition (PECVD), High-Density Plasma (HDP) deposition, or the like.

[0018] The film stack further includes a hard mask layer **108** formed over the ARC **106** and the target layer **102**. The hard mask layer **108** may be formed of a material that comprises a metal (e.g., titanium nitride, titanium, tantalum nitride, tantalum, a metal-doped carbide (e.g., tungsten carbide), or the like) and/or a metalloid (e.g., silicon nitride, boron nitride, silicon carbide, or the like), and may be formed by PVD, Radio Frequency PVD (RFPVD), CVD, Atomic Layer Deposition (ALD), or the like. In subsequent processing steps, a pattern is formed on the hard mask layer **108** using an embodiment patterning process. The hard mask layer **108** is then used as an etching mask for etching the target layer **102**, where the pattern of the hard mask layer **108** is transferred to the target layer **102**.

[0019] The film stack further includes a dielectric layer **110** formed over the hard mask layer **108**. The dielectric layer **110** may be a silicon oxide, and may be formed from borophosphosilicate tetraethylorthosilicate (BPTEOS) or undoped tetraethylorthosilicate (TEOS), or formed by CVD, ALD, spin-on coating, or the like. In some embodiments, the dielectric layer **110** acts as an etch stop layer for patterning a bottom layer of a bi-layer or tri-layer photoresist mask. In some embodiments, the dielectric layer **110** also acts as an anti-reflective coating.

[0020] A tri-layer photoresist **120** is formed on the film stack over the dielectric

[0021] layer **110**. The tri-layer photoresist **120** includes a bottom layer **114**, a middle layer **116** over the bottom layer **114**, and an upper layer **118** over the middle layer **116**. The bottom layer **114** and upper layer **118** may be formed of photoresists (e.g., photosensitive materials), which include organic materials. In some embodiments, the upper layer **118** is a material comprising carbon such as an organic material, e.g. diazonaphthoquinone, novolac resins, Bisphenol A novolac epoxy, off-stoichiometry thiol-enes (OSTE) polymer, the like, or a combination thereof. In some embodiments, the bottom layer **114** may also be a bottom anti-reflective coating (BARC) layer.

[0022] The middle layer **116** may comprise an inorganic material, which may be a nitride (such as silicon nitride), an oxynitride (such as silicon oxynitride), an oxide (such as silicon oxide), combinations of these or the like. The middle layer **116** may have a high etching selectivity relative to the upper layer **118** (after modification as described below) and the bottom layer **114**. The various layers of the tri-layer photoresist **120** may be blanket deposited sequentially using, for example, spin-on processes. Other suitable deposition process may be used, such as CVD, ALD, PVD, or the like.

[0023] Although a tri-layer photoresist **120** is discussed, in some embodiments, the photoresist **120** may be a monolayer or a bilayer (e.g., comprising only the bottom layer **114** and the upper layer **118** without the middle layer **116**) photoresist. The type of photoresist used (e.g., monolayer, bilayer, or tri-layer) may depend on the photolithography process used to pattern the target layer **102** or mandrel/spacer layer overlying the dielectric layer **110**.

[0024] The upper layer **118** is patterned using a photolithographic process, as discussed in greater detail below.

Subsequently, the upper layer **118** is used as an etching mask for patterning of the middle layer **116** (see e.g., FIG. 4). The middle layer **116** is then used as an etching mask for patterning of the bottom layer **114** (see e.g., FIG. 5). The bottom layer **114** is then used to pattern the dielectric layer **110** and the hard mask layer **108** (see FIGS. 6-7), the hard mask layer **108** is then used to pattern the ARC **106** (see FIG. 8) and the target layer **102** (see FIG. 9).

[0025] The upper layer **118** is patterned using any suitable photolithography process to form openings **122** therein. To achieve narrow line widths directly in a single patterning process, EUV may be used as a light source in the photolithography process. As an example of patterning openings **122** in the upper layer **118**, a photomask (not shown) may be disposed between a radiation beam and the upper layer **118**. The upper layer **118** may then be exposed to the radiation beam. In some embodiments, the radiation beam may include a radiation source, such as EUV, an ultraviolet (UV) or an excimer laser such as a 248 nm beam from a Krypton Fluoride (KrF) excimer laser, a 193 nm beam from an Argon Fluoride (ArF) excimer laser, or a 157 nm beam from a F2 excimer laser, or the like while the photomask masks areas of the upper layer **118**. Exposure of the upper layer **118** may be performed using an immersion lithography system to increase resolution and decrease the minimum achievable pitch. A bake or cure operation may be performed to harden the upper layer **118**, and a developer may be used to remove either the exposed or unexposed portions of the upper layer **118** depending on whether a positive or negative resist is used.

[0026] The openings **122** may have strip shapes in a plan view (not illustrated). The pitch P_1 of the openings **122** may be the minimum pitch achievable using photolithographic processes alone. For example, in some embodiments, the pitch P_1 of the openings **122** is in a range of 30 nm and 100 nm. Other pitches P_1 of the openings **122** are also contemplated. The width W_1 of the openings **122** may be in a range of 15 nm and 50 nm, and the width W_2 of discrete remaining portions of the upper layer **118** may be in a range of 15 nm and 50 nm. Other values of W_1 and W_2 are also contemplated.

[0027] In FIG. 2, the upper layer **118** is treated with a treatment process **200**, which converts the upper layer **118** to a treated upper layer **220**. The treatment process **200** may lead to a reduction in line width roughness (LWR) of subsequently formed conductive features in the target layer **102** (see below, FIGS. 9-10), such as metal lines. Resulting smoother line profiles may improve device performance, such as by decreasing sheet resistance. In some embodiments, a de-scum process (as described below in, e.g., FIG. 3) may be performed prior to the treatment process **200**.

[0028] In some embodiments, the treatment process **200** is a plasma treatment, such as with an O_2 plasma. The O_2 plasma may react with the organic photoresist material of the upper layer **118**, converting the upper layer **118** to the treated upper layer **220**. In some embodiments, the modification of the upper layer **118** to the treated upper layer **220** is an oxidation of the carbon-containing exposed surfaces of the upper layer **118**. The oxidation of the edges and profile of the treated upper layer **220** may modify bonds on the exposed surfaces of the treated upper layer **220** and provide improved etching selectivity with the material of the middle layer **116** compared to the untreated upper layer **118**, which may improve the etching of the middle layer **116** in a subsequent

step (see below, FIG. 4). This may lead to reduced LWR of subsequently formed conductive features in the target layer **102**.

[0029] The treatment process **200** may be performed at a temperature less than 200° C., such as in a range of 70° C. to 200° C., which is advantageous for improving etching selectivity of the treated upper layer **220** with respect to the middle layer **116**. Performing the treatment process **200** at a temperature less than 70° C. may be disadvantageous by decreasing efficiency of the plasma treatment, which may lead to worse etching selectivity of the treated upper layer **220** with the middle layer **116**. Performing the treatment process **200** at a temperature greater than 200° C. may be disadvantageous by damaging the profile of the treated upper layer **220**, which may lead to increased LWR of subsequently formed conductive features in the target layer **102**.

[0030] The treatment process **200** may be performed for a duration in a range of 1 second to 10 seconds, which is advantageous for improving etching selectivity of the treated upper layer **220** with the middle layer **116**. Performing the treatment process **200** for less than 1 second may be disadvantageous by decreasing efficiency of the plasma treatment, which may lead to worse etching selectivity of the treated upper layer **220** with the middle layer **116**. Performing the treatment process **200** for more than 10 seconds may be disadvantageous by damaging the profile of the treated upper layer **220**, which may lead to increased LWR of subsequently formed conductive features in the target layer **102**.

[0031] FIG. 3 illustrates a de-scum process **128** on the semiconductor device **100** after the patterning of the upper layer **118** as illustrated in FIG. 1. The de-scum process **128** uniformly removes a small amount of material from the exposed parts of the treated upper layer **220** and middle layer **116**. The de-scum process **128** provides vertical resist profiles of the treated upper layer **220** and removes residue of the treated upper layer **220** which remain in openings **122** after patterning. The de-scum process may use a wet or dry etch process to facilitate the removal of material of the treated upper layer **220** and middle layer **116**. Suitable chemical etchants for a wet de-scum process include tetramethylammonium hydroxide (TMAH), sulfuric acid (H_2SO_4), potassium hydroxide (KOH), boron hydroxide (BOH), other suitable acids or hydroxides, or combinations thereof. Suitable process gases for a dry de-scum process include oxygen, a fluorine containing gas, a chlorine containing gas, an O_2 based gas, or combinations thereof. The de-scum process **128** may comprise one or more process steps. As a result of the de-scum process **128**, openings **122** are enlarged into openings **124**. The width W_3 of the openings **124** may be in a range of 16 nm and 53 nm. Correspondingly, the width W_4 of portions of the treated upper layer **220** at its narrowest points may be in a range of 12 nm and 49 nm. Although FIGS. 2 and 3 describe the de-scum process **128** being performed after the treatment process **200**, in other embodiments, the de-scum process **128** may be performed prior to the treatment process **200**.

[0032] After the patterning of the upper layer **118**, the treatment process **200**, and the de-scum process **128**, FIG. 4 illustrates the transferring of the pattern of the treated upper layer **220** after conversion to the middle layer **116** in an etching process. The etching process is anisotropic, so that the openings **124** are extended through the middle layer **116**

and have about the same sizes in the middle layer 116 as they do in the overlying layer. The oxidized top surfaces and sidewalls of the treated upper layer 220 may be resistant to damage from the etching process, which may reduce the amount of the treated upper layer 220 consumed by the etching process. Because sufficient amounts of the treated upper layer 220 are not consumed by the etching process, the pattern of the treated upper layer 220 is transferred to the middle layer 116 by the etching process with fewer defects. The pattern of the middle layer 116 having fewer defects may lead to reduced roughness of the sidewalls of the openings 124 and improved subsequent transfer of the pattern to lower layers, which may reduce LWR of subsequently formed conductive features in the target layer 102.

[0033] In FIG. 5, an etching process is performed to transfer the pattern of the middle layer 116 to the bottom layer 114, thereby extending the openings 124 through the bottom layer 114. The etching process of the bottom layer 114 is anisotropic, so that the openings 124 in the middle layer 116 are extended through the bottom layer 114 and have about the same sizes in the bottom layer 114 as they do in the middle layer 116. The oxidized top surfaces and sidewalls of the treated upper layer 220 may reduce the amount of the treated upper layer 220 consumed while transferring the pattern of the treated upper layer 220 to the middle layer 116, which may reduce roughness of the sidewalls of the openings 124 and transfer the pattern of the middle layer 116 to the bottom layer 114 with fewer defects. The pattern of the bottom layer 114 having fewer defects may lead to improved subsequent transfer of the pattern to lower layers, which may reduce LWR of subsequently formed conductive features in the target layer 102. As part of etching the bottom layer 114, remaining portions of the treated upper layer 220 may be consumed. In some embodiments, remaining portions of the treated upper layer 220 are removed by a separate process prior to or after the etching of the bottom layer 114.

[0034] FIGS. 6 and 7 illustrate the continuing patterning of subsequent layers down to the target layer 102 by transferring the pattern in overlying layers to the underlying layers. FIG. 8 illustrates depositing features in the target layer 102. FIG. 9 is a plan view following the depositing of features in the target layer 102.

[0035] In FIG. 6, the pattern of the bottom layer 114 is transferred to the dielectric layer 110 and the hard mask layer 108 using an etching process, forming openings 134 through the dielectric layer 110 and the hard mask layer 108. The etching process of the dielectric layer 110 and the hard mask layer 108 is anisotropic, so that the openings 124 in the bottom layer 114 (see above, FIG. 5) are extended through the dielectric layer 110 and the hard mask layer 108, and so that the openings 134 have about the same sizes in the dielectric layer 110 as the openings 124 in the bottom layer 114. The oxidized top surfaces and sidewalls of the treated upper layer 220 may reduce the amount of the treated upper layer 220 consumed while transferring the pattern of the treated upper layer 220 to the middle layer 116, which may lead to a transfer of the pattern to the bottom layer 114 with fewer defects and reduced roughness of the sidewalls of the openings 124. As the pattern of the bottom layer 114 has reduced defects, the transfer of the pattern to the dielectric layer 110 and the hard mask layer 108 may produce fewer defects and lead to reduced roughness of the sidewalls of the openings 134, which may further lead to reduced LWR of

subsequently formed conductive features in the target layer 102. During etching the dielectric layer 110 and the hard mask layer 108, the middle layer 116 is consumed, and the bottom layer 114 may be at least partially consumed. In embodiments when the bottom layer 114 is not completely consumed while etching the dielectric layer 110, an ashing process may be performed to remove remaining residue of the bottom layer 114.

[0036] In FIG. 7, the ARC 106 and target layer 102 are sequentially etched using the hard mask layer 108 as an etching mask to extend openings 134 into the target layer 102 to create openings 140. The oxidized top surfaces and sidewalls of the treated upper layer 220 (see above, FIG. 2) may lead to a transfer of the pattern of the treated upper layer 220 to the hard mask layer 108 with reduced defects and reduced roughness of the sidewalls of the openings 134. This may lead to a transfer of the pattern to the target layer 102 with fewer defects, reduced roughness of the sidewalls of the openings 140 in the target layer 102, and reduced LWR of subsequently formed conductive features in the openings 140. Prior to etching the ARC 106 and the target layer 102, a wet cleaning may be performed to remove any remaining portions of the dielectric layer 110. Etching the target layer 102 may comprise an anisotropic dry etch process and/or a wet etch process, which sequentially etches through the ARC 106 to the target layer 102. After the openings 134 are patterned to form openings 140, a wet cleaning may be performed to remove any remaining portions of the hard mask layer 108 and the ARC 106 (see below, FIG. 8).

[0037] In FIG. 8, after the openings 140 are patterned in the target layer 102, features may be formed in the openings 140. In an embodiment, the target layer 102 is a low-k dielectric, and the patterned target layer 102 provides an IMD for an interconnect structure. Conductive features such as copper lines, copper vias, and/or cobalt plugs may be formed in the IMD layer as illustrated by FIG. 8. Forming the conductive features may include depositing one or more liners 142 along sidewalls and a bottom surface of the openings 140 (see FIG. 7). The liners 142 may comprise TiO, TiN, TaO, TaN, or the like and may provide diffusion barrier, adhesion, and/or seed layers for the conductive feature. The liners may be deposited using any suitable process, such as PVD, CVD, ALD, and the like. The conductive features may have reduced LWR due to the oxidized top surfaces and sidewalls of the treated upper layer 220 (see above, FIG. 2) leading to the pattern of the treated upper layer 220 being transferred more cleanly to the target layer 102.

[0038] After the liners 142 are deposited, remaining portions of the opening may be filled with a conductive material 144, such as copper using, e.g., PVD, plating, or the like. The conductive material 144 may be initially deposited to overfill the openings 140 (see above, FIG. 7), and a planarization process performed to remove excess portions of the conductive material 144 over the target layer 102. Thus, conductive features may be formed in the target layer 102.

[0039] FIG. 9 illustrates a top down view of the semiconductor device 100 after the features have been formed in the target layer 102, in accordance with some embodiments. FIG. 9 is a partial view of a pattern of conductive material 144 surrounded by the target layer 102. Liners 142 may be between the target layer 102 and the conductive material 144.

[0040] FIG. 10 follows from FIG. 1 and shows a film 302 formed over exposed surfaces of the top layer 118 and the middle layer 116. In some embodiments, the film 302 is formed over a treated upper layer 220, as illustrated above in FIG. 2. The film 302 is formed to a larger thickness on top surfaces of the upper layer 118 than on exposed top surfaces of the middle layer 116 due to limitations of the deposition process and the relatively high aspect ratio of the openings 122 between portions of the patterned top layer 118. The relatively thicker portions of the film 302 on the top surfaces of the upper layer 118 protect the profile of the top layer 118 during a subsequent etching process (see below, FIG. 12), which improves etching selectivity between the top layer 118 and the middle layer 116 and reduces LWR in subsequently formed conductive features such as in the target layer 102. In some embodiments, the film 302 comprises silicon oxide, silicon nitride, silicon carbide, silicon oxycarbide, silicon oxynitride, silicon oxycarbonitride, the like, or a combination thereof. The film 302 may be formed using Atomic Layer Deposition (ALD), Plasma Enhanced Atomic Layer Deposition (PEALD), CVD, PVD, the like, or a combination thereof. In some embodiments, the film 302 is silicon oxide and is formed with PEALD using oxygen, tris (dimethylamino) silane (3DMAS), bis (tertiary-butylamino) silane (BTBAS), bis (diethylamino) silane (BDEAS), the like, or a combination thereof as precursor gases.

[0041] The film 302 may be formed to a thickness T_1 in a range of 3 Å to 30 Å on top surfaces of the upper layer 118, which is advantageous for protecting the profile of the upper layer 118 during a subsequent etching process. Forming the film 302 to a thickness less than 3 Å on top surfaces of the upper layer 118 may not protect the profile of the top layer 118 during the subsequent etching process, leading to increased LWR of subsequently formed conductive features of the target layer 102 (see above, FIG. 8). Forming the film 302 to a thickness greater than 30 Å on top surfaces of the upper layer 118 may lead to insufficient etching of the middle layer 116 during the subsequent etching process, leading to poorly formed conductive features of the target layer 102.

[0042] The film 302 may be formed to a thickness T_2 in a range of 5 Å to 25 Å on top surfaces of the middle layer 116. In some embodiments, a ratio of the thickness T_1 to the thickness T_2 is in a range of 1 to 1.5, which is advantageous for protecting the profile of the upper layer 118 during a subsequent etching process. Forming the film 302 with a ratio of $T_1:T_2$ less than 1 may not protect the profile of the top layer 118 during the subsequent etching process, leading to increased LWR of subsequently formed conductive features of the target layer 102 (see above, FIG. 9). Forming the film 302 with a ratio of $T_1:T_2$ greater than 1.5 may lead to insufficient etching of the middle layer 116 during the subsequent etching process, leading to poorly formed conductive features of the target layer 102.

[0043] FIG. 11 follows from FIG. 10 and illustrates a thinning of the film 302 by a suitable process. The thinning of the film 302 exposes top surfaces of the middle layer 116 in preparation for a subsequent transfer of the pattern of the upper layer 118 to the middle layer 116 (see below, FIG. 12). The film 302 may be thinned by an anisotropic etch, such as a reactive ion etching (RIE) process. After the thinning process, the film 302 may cover top surfaces and sidewalls of the upper layer 118, which increases the effective height

of the upper layer 118. The higher effective height of the upper layer 118 may lead to a transfer of the pattern of the upper layer 118 to the middle layer 116 with fewer defects.

[0044] The film 302 may have a thickness T_3 in a range of 0.1 Å to 12.5 Å on top surfaces of the upper layer 118 after the thinning process, which is advantageous for protecting the profile of the upper layer 118 during a subsequent etching process. The film 302 having a thickness less than 0.1 Å on top surfaces of the upper layer 118 after the thinning process may not protect the profile of the top layer 118 during the subsequent etching process, leading to increased LWR of subsequently formed conductive features of the target layer 102 (see above, FIG. 9). The film 302 having a thickness greater than 12.5 Å on top surfaces of the upper layer 118 after the thinning process may lead to poor pattern transfer to the middle layer 116 during the subsequent etching process, leading to poorly formed conductive features of the target layer 102.

[0045] FIG. 12 follows from FIG. 11 and illustrates the transferring of the pattern of the upper layer 118 to the middle layer 116 in an etching process. The etching process is anisotropic, such as a reactive ion etching (RIE) process, so that the openings 124 are extended through the middle layer 116 and have about the same sizes in the middle layer 116 as they do in the overlying layer, in accordance with some embodiments. In some embodiments, the thinning of the film 302 and the transferring of the pattern of the upper layer 118 to the middle layer 116 are performed by the same etching process or a separate etching process. The remaining portions of the film 302 on the top surfaces of the upper layer 118 increase the effective height of the upper layer 118 during the etching process. The higher effective height of the upper layer 118 may increase the etch selectivity of the upper layer 118 with the middle layer 116. This can lead to a transfer of the pattern of the upper layer 118 to the middle layer 116 with fewer defects and to subsequent transfers of the pattern to lower layers (see above, FIGS. 5-7) with fewer defect, which can lead to a reduction in line width roughness (LWR) of subsequently formed conductive features in the target layer 102 (see above, FIGS. 8-9), such as metal lines. The etching process may remove portions of the film 302 on top surfaces of the upper layer 118. In some embodiments, portions of the film 302 remain on sidewalls of the upper layer 118 after the etching process, and the patterned portions of the middle layer 116 have larger widths than the patterned portions of the upper layer 118. The remaining portions of the film 302 may be removed in a subsequent etching process of the bottom layer 114 (see FIG. 5, above).

[0046] After the transferring of the pattern of the upper layer 118 to the middle layer 116, subsequent processing steps may proceed as described above in respect to FIGS. 5-9. Due to the film 302 protecting the profile of the upper layer 118, the pattern of the upper layer 118 may be transferred through intermediate layers (e.g. the middle layer 116, the bottom layer 114, the dielectric layer 110, the hard mask 108, and the ARC 106) to the target layer 102 with reduced defects, which may lead to conductive features formed in the target layer 102 having reduced LWR. The resulting smoother line profiles may improve device performance, such as by decreasing sheet resistance.

[0047] By employing the embodiments discussed herein, conductive lines having spacing of 50 nm or less may be reliably formed with fewer manufacturing defects and increased yield in a single patterning process. In other

embodiments where the target layer 102 is a conductive layer or a semiconductor layer, a reverse pattern of the hard mask layer 108 (see above, FIG. 6) may be patterned in the target layer 102 using a similar process as described above with respect to FIG. 7. For example, an additional hard mask (not shown) may be deposited around the hard mask layer 108. The hard mask layer 108 is then removed, and the additional hard mask is used to pattern the target layer 102. The resulting patterned target layer 102 has a negative pattern than the pattern of the hard mask 108.

[0048] Embodiments may achieve advantages. Treating a patterned photoresist with a treatment process (e.g., a plasma treatment) or forming a film over the patterned photoresist can protect the profile of the patterned photoresist during a subsequent etching process. The protected profile of the patterned photoresist layer may lead to a transfer of the pattern of the patterned photoresist to lower layers with reduced defects, which may reduce line width roughness (LWR) of subsequently formed conductive features in a target layer, such as metal lines. Device performance can be improved by the decrease in sheet resistance provided by smoother profiles of the conductive features formed in the target layer.

[0049] In accordance with an embodiment, a method of forming a semiconductor device includes: patterning a photoresist layer on a mask layer; performing an oxygen plasma treatment on the photoresist layer, the oxygen plasma treatment converting the photoresist layer to a treated photoresist layer by oxidizing an exposed surface of the photoresist layer; patterning the mask layer using the treated photoresist layer as a mask; and patterning a target layer using the patterned mask layer as a mask. In an embodiment, the oxygen plasma treatment is performed at a temperature less than 200° C. In an embodiment, the oxygen plasma treatment is performed for a duration in a range of 1 second to 10 seconds. In an embodiment, the photoresist layer includes carbon. In an embodiment, the method further includes performing a de-scum process on the semiconductor device after patterning the photoresist layer. In an embodiment, the de-scum process is performed before performing the oxygen plasma treatment on the photoresist layer. In an embodiment, the de-scum process is performed after performing the oxygen plasma treatment on the photoresist layer.

[0050] In accordance with another embodiment, a method of forming a semiconductor device includes: forming a film over a patterned top layer, the patterned top layer being on a mask layer, the mask layer being over a target layer, wherein openings extend through the patterned top layer to expose the mask layer, a first portion of the film over the patterned top layer has a first thickness, a second portion of the film on a bottom surface of the openings has a second thickness, and the first thickness is greater than the second thickness; patterning the mask layer using the patterned top layer as a mask; and using the patterned mask layer as a mask to pattern the target layer. In an embodiment, the film is silicon oxide, silicon oxycarbide, or silicon oxycarbonitride. In an embodiment, forming the film includes a plasma enhanced atomic layer deposition process. In an embodiment, the method further includes performing a thinning process on the film before patterning the mask layer. In an embodiment, the film has a thickness in a range of 3 Å to 30 Å over a top surface of the patterned top layer before the thinning process. In an embodiment, the film has a thickness in a range of 0.1 Å to 12.5 Å over a top surface of the

patterned top layer after the thinning process. In an embodiment, the mask layer includes a film stack over the target layer and a photoresist over the film stack.

[0051] In accordance with yet another embodiment, a method of forming a semiconductor device includes: forming a film stack over a target layer, the target layer being on a substrate; forming a photoresist over the film stack, the photoresist including a bottom layer, a middle layer over the bottom layer, and a top layer over the middle layer; patterning the top layer; forming an oxide layer on exposed surfaces of the patterned top layer; patterning the middle layer using the patterned top layer as a mask; patterning the bottom layer using the patterned middle layer as a mask; patterning the film stack using the patterned bottom layer as a mask; patterning the target layer using the patterned film stack as a mask to form openings through the target layer; and forming conductive features in the openings. In an embodiment, forming the oxide layer includes an O₂ plasma process. In an embodiment, forming the oxide layer includes depositing an oxide film over the patterned top layer. In an embodiment, the oxide film includes silicon and carbon. In an embodiment, the film stack includes an anti-reflective coating, a hard mask over the anti-reflective coating, and a dielectric layer over the hard mask. In an embodiment, the method further includes forming a liner in the openings.

[0052] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

1. A method comprising:

forming a masking structure over a target layer;

patterning a top portion of the masking structure to form a patterned top portion of the masking structure, the patterned top portion including respective top surfaces and sidewalls;

subjecting the patterned top portion of the masking structure to a treatment that increases etch selectivity of the patterned top portion, when exposed to an etch process, relative to etch selectivity of the patterned top portion in the absence of the treatment;

after subjecting the patterned top portion of the masking structure to the treatment, patterning a lower portion of the masking structure, using the patterned top portion of the masking structure as an etch mask, to form a patterned masking structure; and

patterning the target layer using the patterned masking structure as an etch mask.

2. The method of claim 1, wherein forming the masking structure comprises depositing a hard mask layer on the target layer, and depositing a multi-layer photoresist on the hard mask.

3. The method of claim 1, wherein forming the masking structure comprises depositing an anti-reflective coating (ARC) on the target layer, depositing a hard mask on the

ARC, depositing a dielectric layer on the hard mask, and depositing a multi-layer photoresist layer on the dielectric layer.

4. The method of claim 1, wherein the step of subjecting the patterned top portion of the masking structure to a treatment comprises oxidizing exposed surfaces of the patterned top portion of the masking structure.

5. The method of claim 1, further comprising:

depositing a film on the respective top surfaces and sidewalls of the patterned top portion of the masking structure and on top surfaces of the lower portion of the masking structure exposed by the patterned top portion of the masking structure; and

thinning the film over the respective top surfaces of the patterned top portion of the masking structure and over the top surfaces of the lower portion of the masking structure exposed by the patterned top portion of the masking structure, before the step of patterning the lower portion of the masking structure.

6. The method of claim 3, wherein the step of depositing the multi-layer photoresist on the dielectric layer includes depositing a first photoresist layer, depositing an inorganic layer on the first photoresist layer, and depositing a second photoresist layer on the inorganic layer.

7. The method of claim 6, wherein the step of using the patterned top portion of the masking structure as an etch mask includes etching the inorganic layer using the second photoresist layer as an etch mask, and etching the first photoresist layer using the inorganic layer as an etch mask.

8. The method of claim 4, wherein the treatment includes an oxygen plasma process.

9. The method of claim 8, wherein the oxygen plasma process oxidizes carbon-containing surfaces of the top portion of the masking structure to modify bonds on the carbon-containing surfaces.

10. A method comprising:

depositing a multi-layer mask over a target layer, the multi-layer mask including a first layer and a second layer, the first layer being over the second layer, and having a first etch selectivity to a predefined etch process, the second layer having a second etch selectivity to the predefined etch process;

patterning the first layer;

after patterning the first layer, subjecting the first layer to a treatment process to increase the first etch selectivity of the first layer to a third etch selectivity to the predefined etch process that is greater than the first etch selectivity and great than the second etch selectivity; patterning the second layer using the first layer as an etch mask;

and patterning the target layer using the second layer as an etch mask.

11. The method of claim 10, wherein the step of subjecting the patterned first layer to a treatment process includes exposing the patterned first layer to an oxygen plasma process.

12. The method claim 10, further comprising depositing a film on the patterned first layer before the step of patterning the second layer using the first layer as an etch mask.

13. The method of claim 12, wherein the step of depositing the film on the patterned first layer includes depositing the film to a first thickness on top surfaces of the patterned first layer and depositing the film to a second thickness less than the first thickness of top surfaces of the second layer that are exposed by the patterned first layer; and

thinning back to the film to a third thickness less than the first thickness on top surfaces of the patterned first layer.

14. The method of claim 13, wherein the step of thinning back the film to the third thickness removes the film from over the top surfaces of the second layer exposed by the patterned first layer.

15. The method of claim 13, wherein a ratio of the first thickness to the second thickness is in a range of from 1 to 1.5.

16. A method comprising:

providing a substrate;

forming a patterning structure over the substrate, the patterning structure including an anti-reflective coating, a hard mask over the anti-reflective coating, and a multi-layer photoresist over the hard mask;

treating a top layer of the multi-layer photoresist to increase an etch selectivity of the top layer to a predetermined etch process to a value that is greater than an etch selectivity prior to the treating;

patterning the top layer of the multi-layer photoresist;

processing a lower layer of the multi-layer photoresist using the patterned top layer of the multi-layer photoresist as a process mask; and

processing the hard mask and the anti-reflective coating using the processed lower layer of the multi-layer photoresist as a process mask.

17. The method of claim 16, wherein the step of processing the lower layer of the multi-layer photoresist includes etching a pattern in the lower layer of the multi-layer photoresist, and further wherein the step of processing the hard mask and the anti-reflective coating includes etching the pattern in the hard mask and the anti-reflective coating.

18. The method of claim 16, wherein the step of treating a top layer of the multi-layer photoresist includes subjecting the top layer of the multi-layer photoresist to an oxygen containing plasma.

19. The method of claim 16, wherein the step of treating a top layer of the multi-layer photoresist is performed after the step of patterning the top layer of the multi-layer photoresist.

20. The method of claim 16, further comprising forming a protective film over the patterned top layer of the multi-layer photoresist before the step of processing the lower layer of the multi-layer photoresist.

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